Unintended Consequences of Constraining Discretion: Evidence from Criminal Sentencing

Spencer Cooper* & Andy Ye Yuan†

Abstract

We study how legal actors respond when procedural reforms constrain their discretion, exploiting a Supreme Court decision that raised the evidentiary burden for sentencing enhancements. Enhancement rates fall by nearly 30 percent, yet average sentences rise by roughly two months rather than mechanically declining. Across thirteen charging and bargaining outcomes, we find no evidence of prosecutorial adjustment. Instead, judges respond heterogeneously: lenient judges increase sentences while strict judges reduce them, compressing the distribution of punishment. These results show that constraining one margin of discretion can produce unintended consequences as legal actors adapt along the discretion that remains.

* Assistant Professor of Economics, University of Connecticut, spencer.cooper@uconn.edu
† Assistant Professor of Law, University of Florida Levin College of Law, yuan@law.ufl.edu

We thank Desmond Ang, Jason Baron, Diego Battiston, Katie Bollman, Remy Levin, David Rosenkranz, Steve Ross, David Simon, Michal Šoltés, and numerous conference participants for their helpful comments and advice.

1. Introduction

How do institutional actors respond to reforms that constrain their discretion? Policies intended to limit discretion often leave other decision margins unchanged, allowing actors to adapt in ways that offset or reshape the policy's intended effects. Understanding these behavioral responses is essential for evaluating legal reforms since their consequences depend not only on the direct mechanical effects of new rules but also on how institutional actors adjust their behavior. In this paper, we study these responses in the context of criminal sentencing, where prosecutors and judges exercise substantial discretion across multiple stages of the legal process.

Measuring these behavioral responses is challenging. Many policies and court decisions are multi-dimensional reforms, affecting multiple aspects of the justice system simultaneously. When multiple policy levers are pulled, isolating the causal source of outcome changes becomes difficult. Moreover, highly salient or consequential reforms may induce changes in case composition, such that prosecutors and judges face systematically different defendants before and after the reform. These two challenges make it difficult to distinguish behavioral responses from selection effects. We address these concerns by exploiting the Supreme Court decision *Blakely v. Washington*, which generated a sharp increase in the evidentiary standard required for sentencing enhancements. This allows us to cleanly identify how prosecutors and judges respond to a procedural change that constrains discretion along the punishment severity margin.

Blakely provides an attractive setting for studying behavioral responses to institutional constraints for two reasons. First, the decision directly constrained a specific punishment margin: the use of sentencing enhancements based on facts that increase the statutory maximum sentence. *Blakely* held that such facts must be found by a jury beyond a reasonable doubt rather than by a judge under the preponderance-of-the-evidence standard. Therefore, the decision both raised the evidentiary burden for enhancements and removed judges' authority to determine enhancement-related facts. Importantly, *Blakely* left the underlying charges, convictions, and other features of sentencing unchanged, allowing us to isolate the consequences of constraining this particular margin of punishment. Because the decision applied only to cases in which enhanced sentences were at issue, its direct mechanical effect should be concentrated among those cases.

Second, the direction of the direct effect on sentence length is straightforward. By making sentencing enhancements more difficult to obtain, *Blakely* should mechanically reduce the frequency with which enhanced sentences are imposed. Because an enhanced sentence is longer than the corresponding non-enhanced sentence, a decline in enhancement use should reduce average sentence lengths if prosecutors and judges otherwise leave their behavior unchanged. Any increase in sentence lengths following the reform must therefore reflect behavioral responses along other margins rather than the direct mechanical effect of the decision.

Our strategy to detect a behavioral response to *Blakely* comes in four steps. First, we measure the direct effect of *Blakely* on enhancement rate using a regression discontinuity design with time acting as the running variable. Second, we consider the average impact of enhancement on sentence length. Third, we combine these two results to create the predicted, no-compensation impact of *Blakely* on sentence length. Finally, we estimate the actual impact of *Blakely* on sentence length using regression discontinuity and compare these effects against the no-compensation prediction. If the difference between these two effects is statistically significant, we consider this evidence that prosecutors and/or judges are changing their behavior in some way to cause an increase in sentence lengths.

We find that *Blakely* caused a decrease in enhancement rate of approximately 30% from the pre-decision mean. By estimating the effects of enhancement on sentence, we predict this reduction in enhancement should decrease overall average sentence length by 0.362 months. Instead, *Blakely* causes an increase of nearly 2 months. The difference between these actual and predicted effects is statistically significant and moves in the opposite predicted direction. We check for any discontinuities in defendant demographics, case density, and enhancement propensity score and find smoothness through the timing of *Blakely*. This indicates that the surprising and significant sentence length effects are not driven by changes in case composition but are instead evidence of a behavioral change in charging, bargaining, or sentencing.

Following this baseline analysis, we turn to identifying potential mechanisms for the surprising and counter-intuitive increase in sentence lengths. To do so, we first present conceptual models which offer predictions of how prosecutors and judges may reallocate across other discretionary margins. We then estimate the impact of *Blakely* across a wide set of prosecutor-related outcomes. Specifically, we consider changes in 13 dependent variables across three main categories: charges, plea bargaining, and sentence outcomes plausibly impacted by prosecutor discretion. These are again estimated using a regression discontinuity design. Next, we turn our attention to judges. To consider judge effects, we construct a measure of judge leniency based on pre-*Blakely* average sentence lengths. Using this measure, we estimate the heterogeneous effects of *Blakely* across judge types using a difference-in-differences design.

We find no evidence of systematic behavioral responses by prosecutors. Across all 13 prosecutorial outcomes, estimated effects are either small and statistically insignificant or move in the opposite direction of the observed increase in sentence lengths. These null results are also highly informative. The estimated confidence intervals are sufficiently tight to rule out economically meaningful adjustments along the primary prosecutorial margins through which compensation might occur. For example, for the proportion of index crimes, a key measure of charging severity, the 95 percent confidence interval rules out increases larger than 1.3 percentage points (3.3 percent relative to the pre-*Blakely* mean). Likewise, for plea bargaining outcomes, the confidence intervals rule out more than modest increases in prosecutorial severity, suggesting that prosecutors did not respond to the loss of enhancements by systematically pursuing harsher charges or plea agreements. Importantly, the absence of charging and bargaining responses does not imply that prosecutors were unaffected by *Blakely*. We estimate a 6.5 percent decline in case duration, consistent with the reform reducing litigation over sentencing enhancements. Thus, prosecutors appear to have adapted to the procedural change by resolving cases more quickly rather than by increasing punishment.

We then turn our attention to judges. We first show that cases proceeding to trial, where judges have greater scope to directly influence sentencing outcomes, exhibit an even larger discontinuous increase in sentence lengths following *Blakely*. We next consider heterogeneity in judicial response. Comparing direct effects by judicial leniency types, we find that cases assigned to the most lenient judges exhibit little change in enhancement use, yet account for nearly all of the increase in sentence length after *Blakely*. These patterns stand in sharp contrast to those for strict judges, who experience substantial declines in both enhancements and sentence lengths. As a result, *Blakely* compresses the distribution of sentences across judges, reducing inter-judge sentencing variance.

The heterogeneous response reflects differences in the remaining discretion available to judges. After *Blakely*, judges respond by increasing sentences within the applicable guideline range. But before *Blakely*, strict judges were substantially more likely to impose sentences at the guideline maximum. Thus, strict judges have less scope to increase sentences along the remaining punishment margin than lenient judges. Consistent with this interpretation, we find that the increase in sentence lengths is concentrated among cases with a high pre-*Blakely* probability of receiving an enhancement, with little evidence of spillover to cases where enhancements were unlikely. Together, these findings suggest that judges respond to the loss of enhancement authority by adjusting sentences along remaining discretionary margins, with the magnitude of the response determined by both the discretion available to the judge and the relevance of the lost enhancement margin to the case.

1.1. Contributions to Literature

Our findings contribute to a broad literature examining how organizations and institutional actors respond to changes in incentives and constraints. A common theoretical framework for these responses is Holmström and Milgrom (1991), in which restricting one dimension of performance induces substitution toward others. Empirical work documents such behavioral adjustment in settings ranging from education to health care (Jacob and Levitt 2003; Shearer, Somé, and Fortin 2018; Alexander 2020). Among these, our paper relates most closely to Alexander (2020), which shows that changing bonus qualifications for doctors has the unintended effect of inducing doctors to admit heathier patients. Similarly, we find that constraining one margin of punishment produces an unintended behavioral response along another margin, as judges increase sentence lengths following *Blakely*. Related work shows that changes in institutional incentives can generate significant behavioral responses within public institutions. Khan et al. (2019) find that changes in the incentive structure for tax inspectors lead to significant increases in productivity, while Bertrand et al. (2020) show that civil servants in India adjust their performance in response to constraints on salary growth.

We extend this literature to a setting in which institutional constraints operate not through monetary incentives but by directly limiting an actor's available discretion. In the criminal justice system, *Blakely* constrained judges' ability to impose punishment through sentencing enhancements while leaving other sentencing margins intact. We show that the extent of behavioral adjustment depends on the discretion remaining along those margins: judges with greater scope to increase sentences within the guideline range exhibit the largest responses. Thus, our findings demonstrate that the consequences of institutional constraints depend not only on the constraint itself, but also on the alternative margins available to institutional actors.

Second, we contribute to the literature on prosecutorial and judicial discretion. Existing work shows that prosecutorial decisions affect both conviction outcomes and sentence lengths, and that discretion can either amplify or mitigate disparities depending on institutional context (Rehavi and Starr 2014; Didwania 2022; Agan, Doleac, and Harvey 2023; Shaffer 2023; Shaffer and Harrington 2025; Yuan and Cooper 2025; Sloan 2026; Tuttle 2026). Both empirical and theoretical work also suggests that prosecutors may substitute across margins when incentives or constraints change (Boylan 2005; Silveira 2017). Importantly, this literature does not provide a clear prediction about how prosecutors should respond to a constraint on one punishment margin: prior studies find both increases and decreases in charging severity depending on the institutional environment (Bjerk 2005; Tuttle 2026; Cooper 2026).

Against this backdrop, we provide evidence on whether prosecutors reallocate discretion when a major punishment margin is constrained. We find remarkably little evidence of such reallocation following *Blakely*: across thirteen charging and bargaining outcomes, we find no systematic changes in prosecutorial behavior. The precision of these estimates allows us to rule out economically meaningful responses along several important margins, including changes in charging severity and plea bargaining. At the same time, cases resolve quicker following the decision, indicating that prosecutors were affected by the reform even though they did not respond by shifting punishment toward other observable charging or bargaining margins. These findings show that prosecutorial responses to institutional constraints are not inevitable and may depend on the particular form of discretion that is constrained.

A parallel literature documents substantial heterogeneity in judicial decision-making. Differences across judges are large enough that quasi-random assignment has been widely used as an instrument for punishment severity (Kling 2006; Dobbie et al. 2018; Arnold, Dobbie, and Yang 2018; Lefgren and Leslie 2023).[1] While this literature establishes that judges matter, less is known about the sources and stability of this heterogeneity. Some work links judicial behavior to institutional constraints, reputation, ideology, or learning (Cohen, Klement, and Neeman 2014; Yang 2016; Cohen and Yang 2019; Bhuller and Sigstad 2024; Angelova, Dobbie, and Yang 2026), but provides limited evidence on how procedural rules reshape judicial behavior.

Our findings speak directly to this question. We show that measured judicial heterogeneity can change with institutional design. Following the reform, sentencing outcomes compress across

[1] These are only a small proportion of papers that utilize the "judge fixed effect" instrument design. Frandsen, Lefgren and Leslie (2023) provides a wider, though still incomplete, description of the papers that use this methodology.

judges as sentencing guideline constraints limit the ability of some judges to respond to the decline in enhancement use. This finding is related to a growing literature documenting the use of reference points in judicial sentencing (Leibovitch 2016; Drápal and Šoltés 2024; Burýšková 2026). While this literature shows that judges use features of the sentencing environment—including other cases and sentencing bounds—as reference points when determining punishment, our findings suggest that judges may also use their own pre-reform sentencing patterns as a reference point when institutional rules change. More broadly, our results suggest that judge fixed effects reflect not only persistent differences across judges, but also how those differences interact with the institutional environment in which judges operate.

A separate contribution is that we find little evidence that judges' responses spill over to cases for which the constrained punishment margin was largely irrelevant. The increase in sentence lengths is concentrated among defendants with a high pre-*Blakely* probability of receiving an enhancement, with little response among defendants who were unlikely to receive one. This suggests that judges do not simply respond to the loss of enhancement authority by increasing sentences across cases but instead target their response toward cases for which the lost margin was consequential. This finding relates to evidence that criminal justice decision-makers use case-specific information in exercising discretion. For example, Kuziemko (2013) shows that parole boards make release decisions that are predictive of subsequent recidivism, suggesting that these actors use information about defendants when exercising discretion. Our results provide complementary evidence that judicial responses to institutional constraints can be similarly targeted rather than broadly applied across cases.

Finally, we contribute to the empirical law and economics literature examining institutional changes in criminal sentencing resulting from the Supreme Court's jury-trial jurisprudence. Beginning with *Apprendi v. New Jersey*, the Court decided a series of high-profile cases that strengthened defendants' Sixth Amendment right to a jury trial by requiring prosecutors to prove to a jury, beyond a reasonable doubt, a broader set of facts that affect sentence length. A growing empirical literature examines how these decisions transformed the institutional structure of the criminal justice system (Prescott 2006; Starr & Rehavi 2013; Yang 2015). Most closely related to our work is Starr and Rehavi (2013) examining the impacts of *United States v. Booker*, which expanded federal judges' sentencing discretion by making the Federal Sentencing Guidelines advisory rather than mandatory. They find suggestive evidence that, in the short run, prosecutors responded to increased judicial discretion by more aggressively charging mandatory minimums, the federal equivalent of sentence enhancements. Yang (2015) examines the longer-run effects of *Booker* and finds that the decision increased prosecutors' use of mandatory minimums, particularly against Black defendants, thereby contributing to larger racial disparities in sentencing.

Our work extends this literature in several ways. First, unlike the existing literature, which focuses exclusively on the federal system, our analysis is among the first to examine the effects of these doctrinal developments on state sentencing systems, where the vast majority of criminal prosecutions in the United States occur. Second, the legal intervention we study differs from that considered in prior studies. We examine the effects of *Blakely v. Washington*, which increased the evidentiary burden on prosecutors seeking sentence enhancements in North Carolina. *Blakely* was the precursor to *Booker*, and its most direct effects were felt in state sentencing systems, making the state level the appropriate setting for our analysis.

In addition, our analysis also draws on distinctive features of North Carolina's sentencing regime. Unlike the Federal Sentencing Guidelines, North Carolina's guidelines include aggravated, presumptive, and mitigated sentencing ranges, which allow us to directly measure sentence enhancements. More importantly, *Blakely* did not alter the mandatory nature of the state's sentencing guidelines. As a result, we can credibly identify sentence enhancements in both the pre-*Blakely* and post-*Blakely* periods. Yuan and Cooper (2025) document a large and persistent decline in sentence enhancements immediately following *Blakely*, with the reduction being particularly pronounced among male defendants. This paper extends our prior work. It calibrates the mechanical effect of *Blakely* on sentence length resulting from the decline in sentence enhancements and compares those mechanically predicted changes with the actual changes observed in sentencing outcomes. The discrepancy between mechanical and actual sentencing effects provides the central motivation for our examination of behavioral responses by prosecutors and judges. In contrast, *Booker* did not directly constrain prosecutorial decision making; rather, it altered judicial discretion, thereby indirectly influencing prosecutorial decisions regarding mandatory-minimum charges. Moreover, researchers are unable to calibrate the predictable mechanical sentencing effects of *Booker* in the manner we implement here because of the nature of the federal sentencing guideline and the facts that *Booker* made the guideline non-binding.

The remainder of the paper proceeds as follows. Section 2 describes North Carolina's structured sentencing system and the role of sentence enhancements. Section 3 outlines our empirical strategy and identification approach. In Section 4, we document the direct effects of *Blakely* on sentence enhancements and sentence length, showing that although enhancement use declines sharply, average sentence lengths increase relative to the pre-*Blakely* period. Section 5 examines a wide range of prosecutorial outcomes and finds little evidence of systematic prosecutorial adjustment. In Section 6, we show that the increase in sentence length is instead driven by judicial behavior. We document significant heterogeneity across judges and discuss plausible explanations for the judicial response. Section 7 concludes.

2. Background

2.1. North Carolina Sentencing

The institutional setting for this study is the North Carolina structured sentencing system. Sentencing process begins after conviction, either through plea bargain or trial. The process requires determining minimum sentences, maximum sentences, and final sentences. Under North Carolina's mandatory sentencing guidelines, judges first select a minimum sentence from the applicable guideline range. The corresponding maximum sentence is then mechanically determined as a function of the minimum sentence. Finally, the judge imposes a sentence within the resulting minimum-maximum interval. This paper focuses on the determination of minimum sentences and, in particular, the sentence enhancements that increase the applicable minimum sentence range. Minimum sentences provide a cleaner measure of judicial decision-making than final sentence lengths, which may also reflect factors unrelated to judicial discretion, including defendants' behavior while incarcerated and post-sentencing administrative decisions from other actors, such as parole officers.

Appendix D.1 illustrates North Carolina's structured sentencing grid. The sentencing guidelines are mandatory throughout the sample period. Each defendant is assigned to a specific grid cell based on offense severity and criminal history. Every cell contains three sentencing ranges: a presumptive range, an aggravated range, and a mitigated range. Judges are generally expected to select minimum sentences from the presumptive range. A sentence enhancement occurs when the presence of aggravating factors permits a shift from the presumptive range to the aggravated range. Conversely, mitigating factors may justify a sentence within the mitigated range.

The decision to pursue an aggravated sentence originates with the prosecutor, who must allege the relevant aggravating factors in connection with the charged offense. To obtain an aggravated sentence, the prosecutor must establish the existence of one or more statutory aggravating factors in addition to proving the underlying offense. Some aggravating factors are relatively objective and straightforward to verify, such as whether the defendant possessed a deadly weapon during the commission of the crime. Others are more subjective and potentially more difficult to establish, such as whether the offense was committed in an especially cruel manner. North Carolina General Statute identifies twenty aggravating factors that may support sentence enhancement[2]. A complete list of these factors is provided in the Online Appendix D.2.

Sentence enhancements can substantially increase defendants' expected punishment. The magnitude of the increase varies with offense severity. For example, among defendants with no prior criminal history convicted of a low-level offense, the presumptive minimum sentence range is 4 to 6 months, whereas the aggravated range is 6 to 8 months. In contrast, for a more serious Class C felony, the presumptive range is 58 to 73 months, and the aggravated range is 73 to 92 months. Both the absolute and proportional differences between the presumptive and aggravated ranges tend to increase with offense severity. Consequently, sentence enhancements are particularly consequential in serious cases, where they can increase expected incarceration by several years. This feature of the sentencing guidelines makes enhancements an important margin of prosecutorial and judicial discretion.

2.2. *Blakely v. Washington* and its implication to North Carolina

Our identification strategy exploits the Supreme Court's decision in *Blakely v. Washington*. In *Blakely*, the defendant was convicted of second-degree kidnapping involving domestic violence. Under Washington's sentencing guidelines, prosecutors sought a sentence within the presumptive range of 49 to 53 months. The sentencing judge instead imposed a sentence of 90 months, nearly 70 percent above the upper bound of the presumptive range. The departure was based on the judge's finding that the offense had been committed with "deliberate cruelty," an aggravating factor that authorized a higher sentence. Washington law permitted judges to impose sentences above the standard range whenever "substantial and compelling reasons" justified an exceptional sentence.[3] The central issue in *Blakely* was whether judges could constitutionally increase

[2] § 15A-1340.16.(d)
[3] § 9.94A.120(2)

sentences based on their own finding of facts. In a 5-4 decision, the Supreme Court held that Washington's sentencing scheme violated the defendant's Sixth Amendment right to a jury trial. The Court ruled that any fact increasing the "maximum sentence" must be proven to a jury beyond a reasonable doubt unless admitted by the defendant. The decision was hugely surprising and generated profound impacts on the sentencing guidelines and practices at both the federal and state level (Prescott & Starr 2006).

Blakely substantially altered sentencing procedures in North Carolina. Before the decision, prosecutors could obtain sentence enhancements by proving aggravating factors to a judge under a preponderance-of-evidence standard unless the defendant admitted the relevant facts. After *Blakely*, prosecutors seeking an enhanced sentence were required either to obtain an explicit admission from the defendant or to prove aggravating factors to a jury beyond a reasonable doubt. The decision therefore increased the evidentiary burden associated with sentence enhancements and reduced prosecutors' ability to obtain enhanced sentences based on judicial fact-finding.

North Carolina provides a particularly useful setting for studying the effects of this change because its sentencing guidelines remained mandatory throughout both the pre-*Blakely* and post-*Blakely* periods. As a result, sentence enhancements applied to a clearly defined set of cases and operated through a single institutional channel: prosecutors' ability to establish legally prescribed aggravating factors. This institutional structure allows us to measure changes in enhancement rates cleanly and, more importantly, to quantify both the mechanical and realized effects of enhancements on sentence lengths. Because sentences imposed from the aggravated range are always weakly greater than those imposed from the presumptive range, a decline in enhancement rates mechanically reduces the sentence lengths. Any divergence between the mechanical and realized effects therefore indicates behavioral responses by prosecutors, judges, or both along other margins of decision-making. The remainder of the paper develops this framework and empirically evaluates the extent of such responses.

3. Empirical Framework & Data

To show how *Blakely* creates predictable variation in sentence length, we outline a simple conceptual model. This model illustrates how we obtain a predicted benchmark that we use as a null hypothesis for the effect of *Blakely* on sentence length. We then provide an empirical strategy to show how we estimate these predicted values in practice.

3.1. Empirical Outline

Suppose $\bar{X}_1$ is the average sentence length for defendants with sentence enhancements. $\bar{X}_2$ is the average sentence length for defendants without sentence enhancements. α is the share of defendants with sentence enhancements. The overall average sentence length $\bar{X}$ can be written as

$$\bar{X} = \alpha * \bar{X}_1 + (1 - \alpha) * \bar{X}_2$$

where $\bar{X}$ is an increasing function in α. Note that $\bar{X}_1 - \bar{X}_2 > 0$ by assumption. By this monotonicity assumption, $\bar{X}$ will decrease if α decreases.

Let α_{pre} and α_{post} represent the share of defendants with sentence enhancement pre- and post-*Blakely*. Then the effect of *Blakely* on sentence length, which we denote as $\Delta_{Blakely}$, can be written as follows:

$$\Delta_{Blakely} = \bar{X}_{post} - \bar{X}_{pre}$$
$$= \alpha_{post} * \bar{X}_1 + \left(1 - \alpha_{post}\right) * \bar{X}_2 - \alpha_{pre} * \bar{X}_1 + \left(1 - \alpha_{pre}\right) * \bar{X}_2$$
$$= \left(\alpha_{post} - \alpha_{pre}\right)\bar{X}_1 - \left(\alpha_{post} - \alpha_{pre}\right)\bar{X}_2$$
$$= \left(\alpha_{post} - \alpha_{pre}\right)(\bar{X}_1 - \bar{X}_2)$$

In theory, assuming that *Blakely* does not change the average sentence lengths for defendants with and without sentence enhancements, the effects of *Blakely* on sentence length can be written as above, which has two components: (1) $\alpha_{post} - \alpha_{pre}$, which are the effects of *Blakely* on sentence enhancements, (2) the difference in sentence length between defendants with sentence enhancements and those without sentence enhancements. Each of these parts can be estimated empirically to derive our predicted benchmark, $\widehat{\Delta}_{blakely}$.

3.2. Empirical Strategy

For a case to receive enhancement, prosecutors must explicitly file intent to prove specific aggravating factors at least 30 days before a plea deal is reached or a trial is held.[4] Thus, the prosecutor must choose whether to pursue enhancement or not for each case. For some cases, pursuing enhancement might be obvious if evidence of a specific aggravating factor is strong and

[4] See *N.C. G.S.* §15A-1340.16(a6).

clear.[5] However, for some aggravating factors, such as "The offense was especially heinous, atrocious, or cruel" or the offense created "…damage causing great monetary loss…," the proof of such factors will be more discretionary. For cases of this type, a prosecutor will weigh the cost of proving such factors against the benefit of a potentially higher sentence length. Because *Blakely* increased the burden of proof required to prove aggravating factors, the cost of pursuing enhancement is higher post-*Blakely*. Therefore, *Blakely* is expected to decrease enhancements as the cost for pursuing enhancement for some marginal cases will be too high post-*Blakely.* Because enhancements strictly increase sentence lengths, *Blakely* is expected to decrease average sentence length.

However, enhancements make up only a small proportion of all cases (only 2.88% of cases received enhancement pre-*Blakely* between the years of 2001 and 2004). Thus, any effects on sentence length are likely to be lost in noise created by a large number of cases unaffected by *Blakely.* Our goal is to narrow our sample to those cases that are most likely to receive enhancement. Doing so is difficult; enhancement is primarily based upon the existence of aggravating factors unobserved in our data. We attempt to predict enhancement likelihood using a logistic regression with crime type and criminal history points as predictors:

Equation 3: $$\Pr(enhancement_i = 1 | \, crime\ type_i, history\ points_i) = \mathrm{F}(crime\ type_i + history\ points_i)$$

where F is the cumulative standard logistic distribution function

We then build a propensity score using the logistic regression results. While more regressors could be included, we find that the best prediction comes from this simpler model. For the majority of our analysis, we reduce our sample to cases with a propensity score of 5% or higher. This reduces our sample of cases from 117,222 cases to 27,263 and raises our pre-*Blakely* mean in aggravation from 3.79% to 7.21%. The purpose of this restriction is not to estimate causal treatment effects of enhancement, but to increase statistical power in detecting changes driven by *Blakely*. Because the decision affects only cases plausibly eligible for enhancement, focusing on this subset sharpens

[5] For example, carrying a firearm during the crime or involving an individual under the age of 16 in the crime are relatively clear cut to prove compared to other, more subjective aggravating factors.

the first stage while preserving the interpretation of our estimates. We show in robustness checks that results are qualitatively similar when considering the full sample of cases.

Because enhancements always increase sentence lengths, we can predict the mechanical expected decrease in sentence length following *Blakely.* We do this by estimating the average sentence length increase caused by enhancement, holding other court factors constant, and multiplying this effect by the estimated effect of *Blakely* on enhancement rate. The product of these two estimates provides a benchmark that represents the expected effect of *Blakely* if criminals, prosecutors, and judges do not change their behavior in a way that impacts sentence length (except through this decrease in enhancement). This benchmark serves as a sharp null hypothesis: any observed change in sentence length that differs from this predicted value must reflect behavioral responses by prosecutors or judges along margins other than enhancement use. This process is outlined in Equation 4, which gives our three main models for our econometric analysis:

Equation 4:

(1) $enhancement_{itg} = \beta_0 + \beta_1 Blakely_t + \beta_2 month_t + \beta_3(Blakely \cdot month_t) + \eta_g + u_{it}$

(2) $sentence\_length_{itg} = \delta_0 + \delta_1 enhancement_{it} + \eta_g + u_{it}$

(3) $sentence\_length_{itg} = \gamma_0 + \gamma_1 Blakely_t + \gamma_2 month_t + \gamma_3(Blakely \cdot month_t) + \eta_g + u_{it}$

$$H_0: \quad \gamma_1 = (\beta_1 \times \delta_1)$$
$$H_1: \quad \gamma_1 \neq (\beta_1 \times \delta_1)$$

The dependent variable in the first equation, $enhancement_{ictg}$, is a binary measure that equals one if case i in county c at half-year t sentenced in grid cell g receives enhancement at sentencing. Note that $\beta_1 \times \delta_1$ is equivalent to $\Delta_{blakely}$ in the conceptual model. The first model in Equation 4 estimates the change enhancement rates, such that $\beta_1 = (\alpha_{post} - \alpha_{pre})$ from the conceptual model. Similarly, the second model estimates the average increase in sentence length for enhanced cases over non-enhanced cases, such that $\delta_1 = (\bar{X}_1 - \bar{X}_2)$. This second model is estimated using only pre-*Blakely* cases. The third model gives the observed, actual effect of *Blakely* on sentence length. We then consider how this observed effect differs from the predicted benchmark. If the null hypothesis is rejected, this is evidence of changed behavior from either prosecutors or judges.

This strategy is effectively a regression discontinuity in time analysis. The validity of this method depends upon *Blakely* occurring without other shocks or policies happening at the same time. In our research of North Carolina sentencing procedure, we find no evidence of any relevant changes or policies within the time bandwidths we use. Empirically, we check for major concurrent changes or potential case-timing manipulation through a series of McCrary tests. First, we consider whether case average propensity score is smooth across the *Blakely* timing using both the propensity score restricted sample and the entire sample of cases. We then check for smoothness of case density, enhancement propensity, and other defendant characteristics for the propensity score reduced sample. These smoothness tests are shown in Figure 1 in the main text and Figure A.1 in the online Appendix. All tests show smoothness through the *Blakely* timing with no evidence of any discontinuous changes in the sample composition. To ensure results are not driven by specific bandwidth or kernel choices, we vary bandwidths between 9, 12, and 15 months and consider results under rectangular, triangular, and Epanechnikov kernels. We also show results hold using quadratic fits rather than standard local linear regressions.

Because our running variable is time, one might worry that our results are driven by an anomaly in time trends that happen to occur near the timing of *Blakely.* To ensure we are capturing the causal effects of *Blakely* and not momentary deviations in enhancement or sentence lengths, we perform additional analyses under a standard event study model using a 6-year time frame (3 years before and after the timing of *Blakely*). Graphs and tables for these results are presented in the online appendix but are mentioned in Section 5 to show consistency of our estimated effects.

3.3. Data

We use records provided by the Administrative Office of the Courts of North Carolina that give all post-arrest charges from 2001 to 2007. These records cover the universe of state-level criminal cases in North Carolina during this period, allowing us to observe charging, bargaining, and sentencing decisions for a large and representative population of defendants. Because North Carolina employs a structured sentencing system similar to those used in many other states, our findings are likely to generalize to other jurisdictions that employ structured or guideline-based sentencing regimes. The data contains detailed defendant characteristics and rich case variables throughout the legal process, including at the charging, conviction, and disposition stages. Defendant characteristics include race, sex, criminal history points, age, and the defendant's type of counsel. We observe the specific statutes a defendant is charged with, the statutes at conviction, whether the case was resolved by plea bargain, and the minimum sentence length at disposition. We leverage this detailed charging information to identify whether *Blakely* sentence length effects are driven by changes in prosecutor charging or bargaining behavior. We also observe the sentencing judge, allowing us to distinguish prosecutorial from judicial responses and to examine heterogeneity across judge types.

It is important to note that we do not explicitly measure whether a case receives enhancement or what factors may be driving enhancement petitions from the prosecutor. However, we can measure whether a case receives enhancement by comparing the minimum sentence length against the prescribed guideline sentence windows. As described in Section 2.1, each grid-cell has a prescribed guideline sentence length for cases with and without enhancement. We consider a case to be enhanced if the minimum sentence length at disposition is within the enhanced sentence length range. While we do not directly observe the aggravating factor findings themselves, any misclassification would tend to attenuate estimated effects, biasing against our main results rather than generating spurious findings.

As noted above in Section 4.2, we reduce the full sample to a set of cases that are most likely to be impacted by the *Blakely* decision. This is done using an enhancement propensity score. The primary difference between the reduced and full sample is offense type. In the full sample of cases, the three most common crime types are drugs (33.3% of sample), burglary (12.2%), and larceny (9.5%). Most of these cases are excluded in the reduced sample, reflecting the institutional reality that sentence enhancements are rarely applied in lower-severity drug and property offenses and are therefore unlikely to be directly affected by *Blakely*. The primary crime types in the reduced sample are robbery (24.7% of sample), assault (22.6%), and sexual assault (21%). These crime types, along with arson, homicide, and kidnapping, retain almost all the cases from the full sample. By restricting attention to these cases with meaningful enhancement risk, we focus on the population for which the *Blakely* decision plausibly altered incentives.

Table 1 provides means and standard deviations for the key variables in our data. We provide statistics for both the full sample and the main, propensity score reduced sample. Note that defendant characteristics such as race, sex, age, and defense type are similar across the two samples. Unsurprisingly, cases in the reduced sample have significantly higher sentence lengths, charge severities, criminal histories, incarceration rate, and enhancement rate. The similarity in observable defendant characteristics across samples suggests that the restriction primarily alters offense composition rather than defendant selection. For each sample, over 95% of cases are resolved through a plea deal. Cases in the propensity score-reduced sample have especially long sentence lengths, with an average of 38.85 months.

Our main sample, which includes cases with propensity scores over 5%, includes cases heard by 143 judges across 73 different districts, allowing us to precisely estimate heterogeneity in judicial responses. Case timing is measured using disposition month, which we use as the running variable in our regression discontinuity design. In all of our sentence length analysis, we focus on minimum sentence length, which determines time to parole eligibility and is the margin directly affected by enhancements under North Carolina law.

4. Mechanical Effects and the Sentencing Puzzle

The direct effects of *Blakely* are presented in Table 2 across three separate panels, one for each component of our main models outlined in Equation 4. Across each panel, the first three columns report discontinuity results for the sample of cases with propensity scores greater than or equal to 0.05, while varying the bandwidth across 9, 12, and 15 months. Columns 4 and 5 utilize a 12-month bandwidth but include the full sample and a sample with propensity scores over 0.03, respectively. These specifications show robustness across bandwidth selections and propensity restrictions.

Panel A reports the impact of *Blakely* on enhancement rate. In each specification, enhancement decreases dramatically following *Blakely.* Column 2 gives the result for the preferred specification, which includes a 12-month bandwidth on the 0.05 propensity score reduced sample. The estimate for this model is a decrease of 2.08 percentage points, which is a 29.5% drop from the pre-*Blakely* mean. This discontinuity is illustrated in Figure 1. While the graph shows a negative trend in enhancement pre-*Blakely*, the decrease in enhancement rate is still discontinuous and large. These effects are corroborated in the event study version of the analysis, displayed in the online appendix Table B.1.

This large decrease in enhancement suggests that sentence lengths should decrease following *Blakely.* As discussed in the methodology section, we can create a rough prediction of the magnitude of this decrease by estimating the impact of enhancement on sentence lengths and multiplying these estimates with the above discontinuity results. Panel B reports the association between enhancement and sentence length, using the same specification schedule as in Panel A. The preferred specification returns an estimate of 18.26 months, which is a 47.3% increase over the pre-*Blakely* mean.

Multiplying the enhancement discontinuity from Panel A by the estimated enhancement–sentence length relationship from Panel B yields the benchmark prediction, $\Delta_{blakely}$. For the preferred specification, this is a 0.38 month decrease following *Blakely.* However, the observed sentence length does not follow this predicted decrease. Instead, sentence length increases after the *Blakely* decision by up to 1.996 months. Figure 2 shows the discontinuity for sentence length with the estimated benchmark of -0.38 marked using a dashed orange line. This effect is fairly consistent across our five specifications and within the event study analysis. This increase in sentence length is surprising and motivates our subsequent analysis of prosecutorial and judicial responses. Before turning to those mechanisms, we first assess the robustness of the main discontinuity results.

We assess the robustness of our findings in several ways. First, online appendix Figure A.2 presents the event-study analogue to Figure 2 and again shows no evidence of a decline in sentence lengths following Blakely. We also demonstrate that the estimated discontinuities in both enhancement rates and sentence lengths are robust to alternative kernel functions and a quadratic polynomial specification (Table B.4), as well as to alternative approaches for calculating standard errors (Table B.5).

Second, we examine cases with low enhancement propensity scores. Because these cases have little ex ante likelihood of receiving an enhancement, they provide a useful test of whether the observed sentence length effects are concentrated among cases most directly affected by Blakely. Results in Table B.6 show main effects for cases with propensities below 0.05 and 0.03. We find enhancement rates still decrease, but with much smaller magnitudes, and noisy null impacts for sentence lengths. While estimates are imprecise, they provide little evidence of sentence length increases among low-propensity cases, suggesting that the main effects are concentrated among cases for which enhancements were plausibly relevant.

Finally, we conduct a placebo timing exercise by estimating discontinuities at alternative cutoff dates. Table B.7 reports uniformly small and statistically insignificant discontinuities for both enhancement rates and sentence lengths when the discontinuity is placed at placebo dates before and after Blakely. These results suggest that our findings are not driven by seasonality, unrelated policy changes, or other spurious timing effects.

5. Prosecutorial Response: Evidence of No Compensation

In this section, we consider whether prosecutor decisions can explain the sentence length effects. When *Blakely* increased costs for pursuing enhancement, prosecutors may have responded by changing other charging or bargaining decisions. Thus, the expected decrease in sentence length following *Blakely* may not occur because prosecutors compensate for less enhancement with higher charges. We consider this possibility by measuring *Blakely* effects across three categories of variables for which prosecutors may have direct or indirect impact: charging, bargaining, and sentencing.

We perform similar analyses as above for a host of dependent variables, estimating effects across different bandwidths for both the 0.05 propensity score restricted sample and the full sample. Specifically, we use 9-, 12-, and 15-month bandwidths across each sample for a total of 6 specifications. These are reported in Tables 3-5. We also show findings are consistent under the event study framework. These results are displayed in the online appendix in Figures A.5 and A.6.

5.1. Charging Decisions

If prosecutors respond to the higher burden of proof by substituting toward harsher charges, we would expect to observe an increase in charge severity following Blakely. Across 5 different charge-related measures, we find no evidence of such a change. These measures are the overall charge severity,[6] a binary measure of whether the charged offense was an index crime,[7] the number of charges before and after consolidation,[8] and a binary measure for whether the case was dismissed. Table 3 contains the discontinuity results for each of these charging dependent variables. Note that the dismissal analysis is only considered for the full sample as the dismissed cases lack the data needed to create the propensity score. Across each specification for each dependent variable, the regression results report small and statistically insignificant effect sizes with small standard errors, indicating no changes in charging behavior following *Blakely*. Even

[6] Charge severity is measured simply on a categorical scale from 1-11, with higher numbers representing more severe crimes.
[7] Index crimes include assault, burglary, robbery, sexual assault, larceny, and homicide.
[8] We created two measures for the number of charges, with or without including the charges that are consolidated. See Appendix D for more details on consolidated charges.

when considering the upper limits of 95% confidence intervals for these estimates, effect sizes are close to zero. The largest of these effects would be charge severity, which has a 95% confidence interval upper bound of around 0.2. This is still a small and insignificant impact given mean charge severity is around 5.9. Furthermore, many of the estimated coefficients have signs that correlate with lower sentence length, indicating they would more likely decrease sentence lengths following *Blakely.* The corresponding event study graphs in Panel A of Figures A4 and A5 likewise show no discernable effects. Thus, increased sentence lengths are not occurring due to prosecutors pursuing more counts or more severe charges.

5.2. Plea Bargaining Outcomes

Even though prosecutors are not changing initial charging decisions, they may still adjust to *Blakely* through more stringent bargaining. If prosecutors are less likely to lower charges in a plea deal compared to pre-*Blakely,* it may explain the null sentence length effects. Thus, we perform the event study and discontinuity analysis again to consider whether prosecutors compensate for fewer enhancements with more stringent plea outcomes. Specifically, we check for changes in the probability of a case having a guilty plea, whether a case's guilty plea leads to incarceration, whether a defendant decreases initial felony charges to a lower felony charge through a plea deal, or whether a defendant decreases initial felony charges to a misdemeanor charge through a plea deal. Each of these are measured using a binary variable. We also include a measure of case duration in this analysis. While case duration is likely driven by many factors, changes in bargaining stringency is one of the main ways a prosecutor can impact case durations.

Table 4 presents the discontinuity results. For the four binary measures of bargaining, we find no discernable changes at the timing of *Blakely,* with small effect sizes tightly centered at zero. The only significant effect is a decrease in plea bargains when using the 12-month bandwidth. However, this effect disappears entirely in other bandwidth specifications. Even when considering this effect, the 95% CI rules out impact sizes larger than -0.032, a relatively small effect compared to the pre-*Blakely* mean of 0.95. The event study analyses largely corroborate this finding. However, case duration appears to decrease following the *Blakely* decision. In the propensity score reduced sample, we estimate a decrease of 23.57 to 26.86 days in case duration, a 6.5 to 7.5 percent decrease from the pre-*Blakely* mean. However, case duration is strongly correlated with higher sentence lengths, even when controlling for many case, area, and timing factors. Thus, a decrease in case duration would predict a decrease in sentence length following *Blakely*; instead, we observe the opposite. Thus, plea decision changes also do not explain the increase we see in sentence length post-*Blakely.*

It's worth noting that *Blakely*'s effect on case duration is an important finding in and of itself. By decreasing prosecutors' incentive to pursue enhancement, the average time taken to case resolution falls. While we cannot directly measure at which point in the justice process time is reduced, the most intuitive mechanisms would be in evidence gathering or plea negotiations. If prosecutors wish pursue enhancement, they may work with law enforcement to gather additional evidence of aggravating factors that occurred in connection with the offense. With lower incentive to pursue enhancement, prosecutors may move to negotiations faster. Likewise, bargaining based on enhancement during plea negotiations are likely to decrease with lower enhancement rate, leading to shorter case durations. In either case, *Blakely* effectively reduces the time needed for case resolution.

5.3. Other Sentencing and Case Outcomes

We next consider whether other sentencing practices can explain the sentence length effects of *Blakely.* Changes in other punishment or leniency measures of sentencing may drive our sentence length results and help indicate whether prosecutors react after *Blakely* by reallocating to other types of punishment. We specifically check whether incarceration rates or probation lengths are affected by *Blakely.* We also consider whether *Blakely* reduced mitigation which would also lead to a sentence length increase. Table 5 presents the results. Once again, the effect sizes are small and insignificant, apart from two specifications with probation as the dependent variable. These two exceptions are both for the full sample of cases and their effects do not hold in the event study analysis, which shows tight null effects for probation as seen in Figure A.5. For the propensity score reduced sample, the 95% CI rules out incarceration effects larger than 0.0383. Our preferred specification also shows a noisy decrease in mitigation. Like the effects on plea bargain, this effect disappears entirely when looking at other bandwidth specifications. Across each specification, standard errors once again remain relatively small, indicating null results are most likely truly zero rather than simply noisy. Taken together, the evidence in Section 5 rules out observable prosecutorial channels as an explanation for the post-*Blakely* increase in sentence lengths, pointing instead to judicial behavior as the source of the unexpected increase.

6. Judicial Response and Mechanism

The main results present a surprising finding – the *Blakely v. Washington* decision drastically reduced enhancement rates, but sentence lengths increased rather than decreasing. This divergence from the mechanical prediction indicates that legal actors adjusted their behavior in response to the reform. The preceding section provides numerous checks by which a prosecutor may increase sentence length through charges, plea negotiations, or other sentencing practices; none of these can explain the sentence length increase.

Having ruled out prosecutorial responses, we now show that the post-*Blakely* increase in sentence length is driven by judicial behavior. Judges may impact sentence lengths in several different ways. Most directly, if a case goes to trial, upon conviction the judge determines the sentence length from within the guideline window. The judge has full discretion in choosing the sentence length within this window. For cases settled by a plea deal, the judge must sign off on the agreed upon deal, a key part of which is the agreed upon sentence length. The judge may also assign a sentence length outside of the recommended, agreed upon sentence length in the deal. If this occurs, the prosecution or defense has the right to withdraw the plea and enter negotiations or move towards trial (North Carolina Defender Manual 2018). But even outside of these direct impacts, the assigned trial judge is likely to impact plea negotiations by reputation. That is, prosecutors and defense attorneys bargain in the shadow of the judge, such that they adjust their expectations for a plea outcome based on how harsh or lenient an assigned judge is. This is evidenced in LaCasse and Payne (1999), which finds plea negotiations occur under the shadow of the judge, particularly for cases with minimum sentencing.

How can we detect whether effects are driven by judicial behavioral changes? First, because judges direct court proceedings, judicial effects may be stronger for cases resolved by jury trial. Thus, we begin our judicial analysis by simply repeating the main sentence length discontinuity analysis for jury-trial cases only. Judicial effects may also be determined by considering heterogeneity in our results across judge type. If our main effects are localized to specific judges, this is evidence that judicial discretion is driving the increased sentence lengths, either directly or indirectly. We specifically consider how the discontinuity effects may differ by judge leniency. While much of the law and economics literature uses judge leniency as an instrument, here we simply construct the leniency measure as a tool to consider whether judges drive results rather than prosecutors. Below we describe how we construct the judge leniency measure, then present the heterogeneous results.

6.1. Measuring Judicial Leniency

There are two separate ways we consider judge leniency – by average enhancement rate and by average sentence length. Judges do not directly impact enhancement but may do so indirectly through reputation effects. That is, a prosecutor may be more likely to pursue enhancement in a given case depending on the assigned judge. For each of these leniency measures, we residualize enhancement/sentence length by baseline defendant characteristics, grid fixed effects, and year fixed effects using only pre-*Blakely* data. Residualizing the measures is important; judges who see especially severe crime types or offenders with a large prior history more often than others will look more severe without controls. By controlling for these factors, we effectively compare judges' sentencing and enhancement propensities for cases of a similar type.

This residual measure is then averaged for each judge and standardized to have a mean of 0 and a standard deviation of 1. Given the resulting continuous measure of averages, we then consider leniency by quartiles, with the lowest quartile representing the most lenient judges and the highest quartile representing the most strict ones. That is, the most lenient judges adjudicate cases with the fewest enhancements/lowest sentence lengths, after controlling for offense class, criminal history, defendant characteristics, and unit-invariant variation over time.

The interpretation of the outcome changes depending on which leniency measure is used. Using the enhancement measure gives the heterogeneous impact of *Blakely* on enhancement and sentence length by judges' different propensities to adjudicate cases with enhancement. But the sentence length measure considers heterogeneity by judges' propensity to give a higher sentence length at conviction. These two measures may not necessarily give the same impact – judges who are harsh in sentencing may not be any more likely to allow for enhancements. To check the similarity between these measures, we plot them against each other. The resulting graph, displayed as Figure A.6 in the online appendix, suggests that most judges who are strict in enhancement are also strict in sentencing. Thus, the results are likely to be similar regardless of leniency measure. We present results for analyses using the sentence length measure, and show results are consistent when using the enhancement measure in the online appendix. We favor the sentence length measure because it is more likely to pick up indirect effects. This is because enhancement is rare, meaning prosecution and defense have less observations to create a prediction of enhancement behavior for a judge. On the other hand, sentence lengths are observed in most cases, meaning sentence leniency is likely a better signal for judge leniency when legal actors engage in plea bargaining.

It's worth noting that because we separate by quartile of judge and not of case, it's possible that judges in one quartile see far more cases than in others. Indeed, we find that with both measures of leniency, the number of cases is nearly equal for the bottom three quartiles, but higher for the strictest judges. This is especially true when using the sentence length measure, where the top quartile has around 27 percent more cases than the other quartiles. One might wonder whether judges in each quartile have observably different characteristics. While we do not observe many judge characteristics, we can say that judge tenure is nearly identical across the quartiles.

Because the heterogeneous analysis effectively splits effects across four groups, we consider a larger time horizon of 6 years (3 years pre and post) for the heterogeneous regressions in order to maintain statistical power. Thus, we now employ a difference-in-differences design as outlined in the below equation:

Equation 5:

$$Y_{itg} = \alpha + \beta_1 Blakely_t + \beta_2 leniency\_50_{it} + \beta_3 leniency\_75_{it} + \beta_4 leniency\_100_{it} + \beta_5(Blakely \cdot leniency\_50_{it}) + \beta_6(Blakely \cdot leniency\_75_{it}) + \beta_7(Blakely \cdot leniency\_100_{it}) + \eta_g + \rho_t + u_{it}$$

where Y_{itg} measures enhancement or sentence length for case i in year t, sentenced in guideline grid g. Coefficient β_1 gives the effect of *Blakely* for the most lenient judges. Coefficients β_5, β_6, and β_7 give the differential impact of *Blakely* for each leniency group compared to the most lenient group.

6.2. Heterogeneous Judicial Responses to *Blakely*

We first briefly present the jury-trial discontinuity results before discussing the heterogenous leniency findings. Figure 4 presents the discontinuity in sentence length at the timing of *Blakely* for cases that were resolved by trial. The graph shows a large increase in sentence length immediately following *Blakely*. Table B.8 presents the estimates for the full set of sentence length discontinuity specifications for the jury trial cases, which shows significantly larger magnitudes compared to the main analysis results. While statistical significance is dependent on specification due to small samples, each regression shows a large increase in sentence length. Figure A.7 in the online appendix illustrates the discontinuity for the full sample of cases and the residual sentence length graph. In each case, sentence length increases greatly following *Blakely.* These findings suggest that the post-*Blakely* increase in sentence lengths is driven by judicial behavior, motivating our analysis of how the response varies with judicial leniency.

Table 6 gives the heterogeneous effects of *Blakely* on both enhancement and sentence length across the four judge leniency quartiles. Columns 2 and 3 give the estimates for the enhancement effect of *Blakely*. Judges who are most lenient (based on the residualized, sentence length-leniency score) exhibit a 2.38 to 2.52 percentage point decrease in enhancement rate following *Blakely.* The next two quartiles of judges show no significant difference from the most lenient quartile. However, the strictest quartile of judges shows a decrease that is nearly double the most lenient group. The difference between the most strict and lenient judge enhancement effects is illustrated in Figure 3. Note that *Blakely* completely eliminates the pre-period discrepancy in enhancement rate.

Columns 3 and 4 of Table 6 report the estimates for the sentence length effects. The most lenient judges see significant increases in sentence length, with an estimated 2.23 to 2.68 month increase in average sentence length following *Blakely,* despite enhancement rates falling. The other quartiles show increasingly negative effects, with the second quartile exhibiting small increases following *Blakely,* the third quartile showing null effects, and the most strict group of judges showing significant decreases in sentence length. Figure 4 visualizes these effects for the most lenient and most strict judicial groups. We also show these results are robust to a discontinuity design with similar specifications to the mechanical analysis in Section 5. These RD results are shown in Tables B.10 and B.11 in the online appendix.

6.3. Interpretation and Mechanisms

These results uncover three important findings: (1) the sentence length increase shown in Section 5.1 of this paper is driven by judges, (2) judicial responses are highly heterogeneous: most judges increase sentence lengths following *Blakely*, while the strictest judges reduce them, and (3) these asymmetric responses generate a pronounced compression of sentencing outcomes across judges, narrowing disparities in both sentence lengths and enhancement use. In regard to the first finding, we interpret the scaling of the effect size with judicial leniency as evidence that the increase in sentence lengths is driven by judicial behavior rather than prosecutorial behavior. While we cannot fully rule out unobserved prosecutorial responses, such as more intensive efforts to seek enhancement-related evidence in each case, it is not clear why such behavior would vary systematically with judicial leniency. Furthermore, we repeat the prosecutor analysis in Section 4.2 separately by judicial leniency quartile and again find consistent null effects.

Our second main result is somewhat surprising. One might expect cases assigned to the strictest judges to exhibit the largest increase in sentence length, since these judges should be most directly affected by the loss of enhancement authority. Instead, we find significant decreases in sentence length, which align with the observed decline in enhancement rates. This pattern suggests that judges differ in their ability to respond along the remaining sentencing margin. Following *Blakely*, judges retain discretion to select a sentence within the applicable guideline range. Because we find no evidence of changes in prosecutorial charging behavior, the increase in sentence lengths does not appear to reflect changes in the severity of cases reaching judges. Instead, judges adjust the sentence imposed within the remaining guideline range. Lenient judges have greater scope to increase sentences along this margin, while strict judges are more likely to be constrained by the upper bound of the guideline range.

To assess whether this explanation is consistent with the data, we examine the share of non-enhanced cases in which the imposed sentence equals the maximum guideline sentence. We estimate the following regression:

Equation 6: $$Y_{itg} = \alpha + \beta_1 leniency\_50_{it} + \beta_2 leniency\_75_{it} + \beta_3 leniency\_100_{it} + \eta_g + \rho_t + u_{it}$$

where $Y_{itg}$is an indicator equal to one if the minimum sentence imposed equals the maximum amount permitted under the guidelines. The results indicate that judges in higher strictness quartiles are substantially more likely to sentence at the guideline maximum. Judges in the strictest quartile are 8.69 percentage points more likely to impose the maximum sentence than judges in the most lenient quartile, a 28 percent increase relative to the latter's pre-*Blakely* mean. We repeat this analysis with varied controls and with the full sample and find similar results (Online Appendix Table B.12). These findings suggest that guideline ceilings bind more frequently for strict judges, limiting their ability to offset the decline in enhancement use, while leaving greater scope for lenient judges to increase sentences within the remaining guideline range.

Our third key finding speaks to how changes in procedural constraints reshape the distribution of sentencing outcomes. Consistent with prior work on judicial discretion, restricting discretion through higher standards of proof is expected to compress sentencing outcomes by limiting the ability of some judges to impose especially severe punishments. In our setting, however, this compression arises through two distinct behavioral responses: harsh judges reduce sentence lengths as their ability to rely on enhancements becomes constrained, while more lenient judges increase sentences within the remaining discretionary range. As a result, sentencing outcomes converge toward the middle of the distribution. As illustrated in Figure 4, both lenient and strict judges move toward a common median sentence length following Blakely, generating substantial compression across judges. This compression result illustrates how constraining one margin of judicial discretion can reshape the distribution of outcomes along another, rather than simply reducing punishment.

6.3.1. Does *Blakely* Spill Over to Other Cases?

The evidence above establishes why the response is concentrated among lenient judges: strict judges have less room to increase sentences within the guideline range. We next examine whether the response is concentrated among the relevant cases – those with a high pre-policy propensity of enhancement. If judges are responding to *Blakely* in an effort to maintain pre-reform sentencing patterns, we would expect larger responses among defendants who would have been likely to receive an enhancement before the reform, but little response among defendants for whom enhancements were unlikely.

We repeat the judicial heterogeneity difference-in-differences analysis for cases with a low predicted probability of enhancement. These results are presented in Table B.13. This analysis provides a sharper test of the results in Table A.6, which examines the overall effects of *Blakely* among low-probability cases. Here, we ask whether the heterogeneous sentence-length responses across judges extend broadly across cases or are concentrated among cases for which enhancement was more likely before *Blakely*. The estimates show that heterogeneity across judges largely dissipates among low-probability cases. Even among the judges with the largest sentence-length increases in the full sample, the estimated effects are statistically insignificant and small in magnitude. We therefore find little evidence of spillover to cases that were unlikely to receive an enhancement before *Blakely*. Instead, the results suggest that the judges who increase sentence lengths following *Blakely* do so primarily among cases for which enhancement was a meaningful pre-policy possibility.

6.3.2. Consistency and Information

One potential explanation for the judicial response is that judges have a desire for consistency in sentencing. This interpretation is consistent with a growing literature showing that judges use other cases in their caseloads and statutory reference points when determining sentence lengths (Leibovitch 2016; Drápal and Šoltés 2024; Burýšková 2026). In our setting, judges may use their pre-*Blakely* sentencing patterns as a reference point when determining how to sentence cases affected by the loss of enhancement authority. For cases that they believe would have received an enhancement absent *Blakely*, judges may seek to maintain their pre-reform sentencing levels by increasing sentences within the remaining guideline range. This interpretation also predicts that judicial responses should be concentrated among cases with a high pre-*Blakely* probability of enhancement, rather than extending broadly to cases that were unlikely to receive an enhancement. Consistent with this prediction, the preceding spillover analysis finds little evidence of heterogeneous judicial responses among low-probability cases.

The consistency interpretation does not require judges to increase sentences all the way to the guideline maximum. Judges may instead adjust sentences according to their assessment of how likely a case would have been to receive an enhancement before *Blakely*. This possibility points to a second, potentially complementary mechanism: changes in the information conveyed through the enhancement process. An enhancement filing may provide judges with information about crime severity because prosecutors seeking an enhancement must present evidence of statutory aggravating factors. This information can affect how judges assess the appropriate sentence within the guideline range. Because *Blakely* increased the cost of obtaining an enhancement, prosecutors may have less incentive to gather and present evidence specifically through the enhancement process. If the resulting signal becomes noisier, judges may face greater uncertainty about which cases warrant lower or higher sentences within the guideline range. This could lead judges to rely more heavily on a pooling equilibrium, moving sentences toward the center of the distribution. Such a response is consistent with the compression we observe in our main results, where lenient judges increase sentences while harsher judges reduce them. A simple model describing this information mechanism is presented in online appendix C.

Importantly, this mechanism does not require that prosecutors actually possess or present less information about aggravating circumstances. Evidence that would previously have been included in an enhancement filing may still be gathered and presented elsewhere in the case. Instead, the relevant change may be in how that information is conveyed to or weighted by judges: without the formal enhancement filing, judges may interpret or weight evidence of aggravating circumstances differently. The information mechanism therefore provides one possible explanation for why judges move toward a pooling equilibrium following *Blakely*, while the consistency interpretation provides a separate explanation for why judges respond primarily in cases that were likely to receive an enhancement before the reform.

These mechanisms of consistency and information may not be mutually exclusive. However, whether it is one or both of these mechanisms, the increase in sentence length we see must be driven by a misprediction on the part of the judges. That is, judges assume there are more cases that would have received enhancement pre-*Blakely* than what really would have. Without this misprediction, we would not see an increase in sentence length among the lenient judges but sentence length would have a consistent, smooth trend through the *Blakely* timing. And while harsh judges may also over-predict in this same way, the observed effect is a decrease in sentence length because of the guideline constraints; the method by which the judge would respond to move towards what they think is a pre-*Blakely* consistent sentence length is less available to the strict judge than it is to lenient ones.

6.3.3. Other explanations and details

We also consider alternative judicial channels, such as changes in pretrial detention or trial-related behavior. Prior work shows that pretrial detention can affect sentencing outcomes (Didwania 2020), raising the possibility that judges might indirectly increase sentences through stricter bail decisions. While we cannot directly observe bail in our data, several patterns suggest this channel is unlikely to drive our results. In particular, sentence length increases are present in both trial and plea cases, indicating that changes in trial procedure or trial selection are not the primary mechanism. Taken together, the evidence points to within-guideline sentencing decisions as the dominant source of the judicial response to *Blakely*.

Finally, one might wonder whether results vary across other judicial characteristics outside of leniency. The data provided to us by the North Carolina AOC does not contain any demographic information about these judges but does give first and last name and the year they started as a judge. Using this information, we found pictures for 53 of the 146 judges in our dataset. With these pictures, we inferred judge race (White versus non-White) and sex. While this is a substantial reduction in our sample, the observation count is slightly higher at around 47% of the original as many of the judges we could not find information for are those with relatively low case counts. We find no evidence of heterogeneity across judge race, sex, or tenure for enhancement results and only marginally significant results for tenure, which is positively correlated with leniency. We likewise find that results for leniency hold when including these demographic factors as controls, though significance levels shrink slightly. We also check for political ideology of the judge, but this results in an even smaller sample – only 22 judges. We likewise find no evidence of heterogeneity here. These results are presented in appendix Table B.14. Taken together, the results in Section 6 illustrate how constraining discretion along one institutional margin can reallocate discretion along another, with distributional and efficiency consequences for sentencing outcomes.

7. Conclusion

This paper studies how institutional actors respond to reforms that constrain discretion in the U.S. judicial system. We examine the Supreme Court's decision in *Blakely v. Washington*, which limited one important margin of punishment by increasing the evidentiary burden required for sentence enhancements. Although *Blakely* sharply reduced the use of sentence enhancements, we find that average sentence lengths increased rather than declined. This increase runs counter to the mechanical implications of the sentencing regime and highlights the importance of behavioral responses to procedural reforms.

We show that this result is driven by judicial, rather than prosecutorial, behavior. Across a wide range of charging and bargaining outcomes, we find no evidence of prosecutorial response. Instead, judges respond heterogeneously: stricter judges reduce sentences as enhancements become harder to obtain, while more lenient judges increase sentences within the remaining discretionary range. These opposing responses generate a compression of sentences across judges and raise average sentence lengths overall. This compression arises because guideline ceilings constrain the strictest judges, leaving more room for lenient judges to increase sentences along the remaining discretionary margin. We also find little evidence that these responses spill over to cases with little ex ante likelihood of receiving an enhancement, suggesting that judicial responses are concentrated among cases most directly affected by the reform. Together, these findings indicate that judges adapt to procedural reforms in ways that can offset their intended effects. The heterogeneous responses across judges show that the effects of institutional constraints depend on the discretion that remains available to decision-makers.

These findings illustrate that constraining one margin of discretion need not reduce punishment as intended. Instead, institutional actors may adapt their behavior within the remaining scope of discretion, altering both the level and distribution of punishment. Our findings suggest that measured judicial heterogeneity depends in part on the institutional environment in which judges operate. More broadly, our results illustrate that evaluating institutional reforms requires accounting for both their direct mechanical effects and the behavioral responses they induce. Ignoring these responses can substantially misrepresent the ultimate effects of legal and policy change.

Our analysis focuses on sentencing outcomes and does not directly observe all potential judicial channels, such as pretrial detention decisions. Future work could explore how procedural constraints affect earlier stages of the criminal process or interact with defendant behavior. More generally, examining how different institutional designs shape adaptive responses remains an important direction for understanding the full consequences of legal reform.

References

Agan, Amanda., Jennifer L. Doleac, and Anna Harvey. (2023). Misdemeanor Prosecution. *Quarterly Journal of Economics*, *138*(3).

Alexander, Diane. 2020. "How Do Doctors Respond to Incentives? Unintended Consequences of Paying Doctors to Reduce Costs." *Journal of Political Economy* 128(11): 4046-4096.

Angelova, Victoria, Will Dobbie, and Crystal Yang. 2026. "Algorithmic Discretion and Human Discretion." *Review of Economic Studies* (forthcoming).

Arnold, David, Will Dobbie, and Crystal S. Yang. 2018. "Racial Bias in Bail Decisions." *Quarterly Journal of Economics* 133 (4).

Bertrand, M., Burgess, R., Chawla, A., and Xu, G. (2020). The glittering prizes: Career incentives and bureaucrat performance. *The Review of Economic Studies,* 87(2):626-655.

Bjerk, David. 2005. "Making the Crime Fit the Penalty: The Role of Prosecutorial Discretion under Mandatory Minimum Sentencing." *Journal of Law and Economics* 48 (2).

Boylan, Richard T. 2005. "What Do Prosecutors Maximize? Evidence from the Careers of U.S. Attorneys." *American Law and Economics Review* 7 (2).

Burýšková, Tereza. 2026. "Categorization and Reference-Dependent Decision-Making under Statutory Thresholds: Evidence from a Criminal Law Reform." Working paper.

Cohen, Alma, Alon Klement, and Zvika Neeman. 2014. "Judicial Decision Making: A Dynamic Reputation Approach." *Journal of Legal Studies.* 44: 133-159.

Cohen, Alma, and Crystal S. Yang. 2019. "Judicial Politics and Sentencing Decisions." *American Economic Journal: Economic Policy* 11 (1).

Cooper, Spencer. 2026. "How Much Do Mandatory Minimums Matter?" *Journal of Law, Economics, and Organization* (forthcoming).

Didwania, Stephanie. 2020. "The Immediate Consequences of Federal Pretrial Detention." *American Law & Economics Review* 22 (1): 24-74.

Didwania, Stephanie. 2022. “Gender Favoritism Among Criminal Prosecutors.” *The Journal of Law and Economics* 65(1): 77-104.

Dobbie, Will, Jacob Goldin, and Crystal S Yang. 2018. “The Effects of Pretrial Detention on Conviction, Future Crime, and Employment: Evidence from Randomly Assigned Judges.” *American Economic Review* 108 (2).

Drápal, Jakub, and Michal Šoltés. 2024 “Sentencing decisions around quantity thresholds: theory and experiment.” *Journal of Experimental Criminology.* 20: 1323-1367.

Frandsen, Brigham, Lars Lefgren, and Emily Leslie. 2023. “Judging Judge Fixed Effects: Testing the Identifying Assumptions in Judge Fixed-Effects Designs.” *American Economic Review* 113(1): 253-77.

Holmström, Bengt and Paul Milgrom. (1991). “Multitask principal-agent analyses: Incentive contracts, asset ownership, and job design.” *Journal of Law, Economics, and Organization,* 7 (Special Issue):24-52.

Jacob, Brian A., and Steven Levitt. 2003. “Rotten Apples: An Investigation of the Prevalence and Predictors of Teacher Cheating.” *Quarterly Journal of Economics.* 118 (3): 843-877.

Khan, A. Q., Khwaja, A.I., and Olken, B. A. 2019. Making moves matter: Experimental evidence on incentivizing bureaucrats through performance-based postings. *American Economic Review,* 109(1):237-270.

Kling, Jeffrey R. 2006. “Incarceration Length, Employment, and Earnings.” *American Economic Review*, 96(3): 863-876.

Kuziemko, Ilyana, 2013. “How Should Inmates Be Released From Prison? An Assessment of Parole Versus Fixed Sentence Regimes.” *Quarterly Journal of Economics,* 128(1): 371-424.

Leslie, Emily and Nolan G. Pope. 2017. “The Unintended Impact of Pretrial Detention on Case Outcomes: Evidence from New York City Arraignments.” *The Journal of Law and Economics*, 60(3):529–557.

Leibovitch Adi. 2016. “Relative Judgements” *The Journal of Legal Studies*, 45(2).

Rehavi, M. Marit, and Sonja B. Starr. 2014. “Racial Disparity in Federal Criminal Sentences.” *Journal of Political Economy* 122 (6).

North Carolina Defender Manual. 2018. Vol. 2, Trial. "Ch. 23: Guilty Pleas (The Plea Procedure)." UNC School of Government. Accessed September 3, 2025. https://defendermanuals.sog.unc.edu/sites/default/files/pdf/23.4%20The%20Plea%20Procedure_0.pdf.

North Carolina Administrative Office of the Courts. 2001–2007. *Administrative Records on Criminal Charges and Sentencing*. Raleigh, NC.

Prescott, JJ. “Essays in Empirical Law and Economics.” (Ph.D. Thesis, Massachusetts Institute of Technology, 2006), http://hdl.handle.net/1721.1/37408.

Prescott, JJ, and Sonja Starr. 2006. “Improving Criminal Jury Decision Making After the Blakely Revolution.” *University of Illinois Law Review* 301.

Shaffer, Hannah. 2023. “Prosecutors, Race, and the Criminal Pipeline.” *The University of Chicago Law Review* 90 (7): 1889–1965.

Shaffer, Hannah, and Emma Harrington. 2025. “Brokers of Bias in the Criminal Justice System: Do Prosecutors Compound or Attenuate Racial Disparities Introduced by Police?” Working paper.

Shearer, Bruce, Nibene H. Somé, and Bernard Fortin. 2018. “Measuring Physicians' Response to Incentives: Evidence on Hours Worked and Multitasking.” *IZA Discussion Paper* 11565.

Silveira, Bernardo S. 2017. “Bargaining With Asymmetric Information: An Empirical Study of Plea Negotiations.” *Econometrica* 85 (2).

Sloan, CarlyWill. 2026. “Do Prosecutor and Defendant Race Pairings Matter? Evidence from Random Assignment.” *Journal of Public Economics* (forthcoming).

Starr, Sonja, M. Marit Rehavi. 2013. “Mandatory Sentencing and Racial Disparity: Assessing the Role of Prosecutors and the Effects of Booker.” *Yale Law Journal* 123: 2.

Tuttle, Cody. 2026. “Racial Disparities in Federal Sentencing: Evidence from Drug Mandatory Minimums.” *Review of Economic Studies* (forthcoming).

Yang, Crystal S. 2015. “Free at Last? Judicial Discretion and Racial Disparities in Federal Sentencing.” *The Journal of Legal Studies* 44 (1): 75-11.

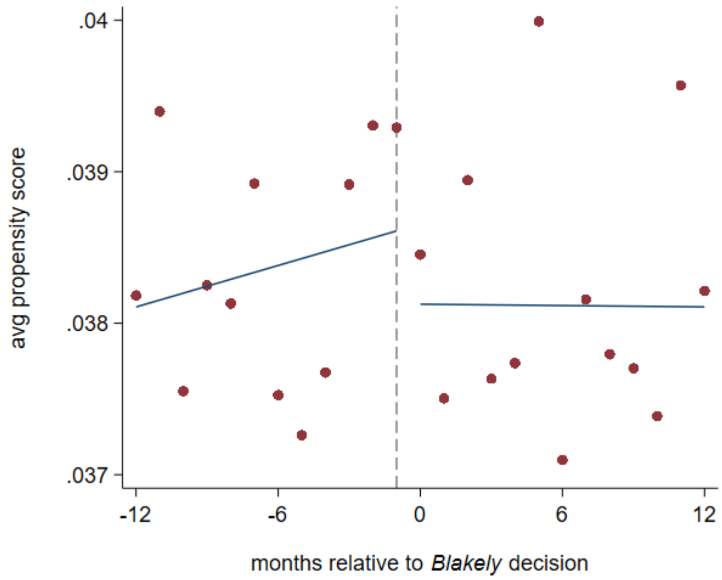
Yang, Crystal S. 2016. “Resource Constraints and the Criminal Justice System: Evidence from Judicial Vacancies.” *American Economic Journal: Economic Policy* 8 (4).

Yuan, Andy and Spencer Cooper. 2025. “Prosecutorial Incentives and Outcome Disparities.” Working paper.

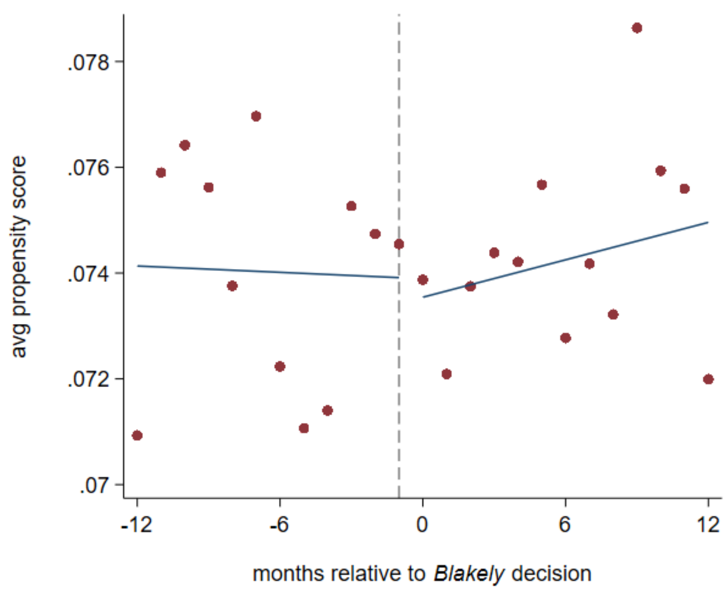
Figures

Figure 1. Smoothness of propensity score

Panel A: *Full sample*

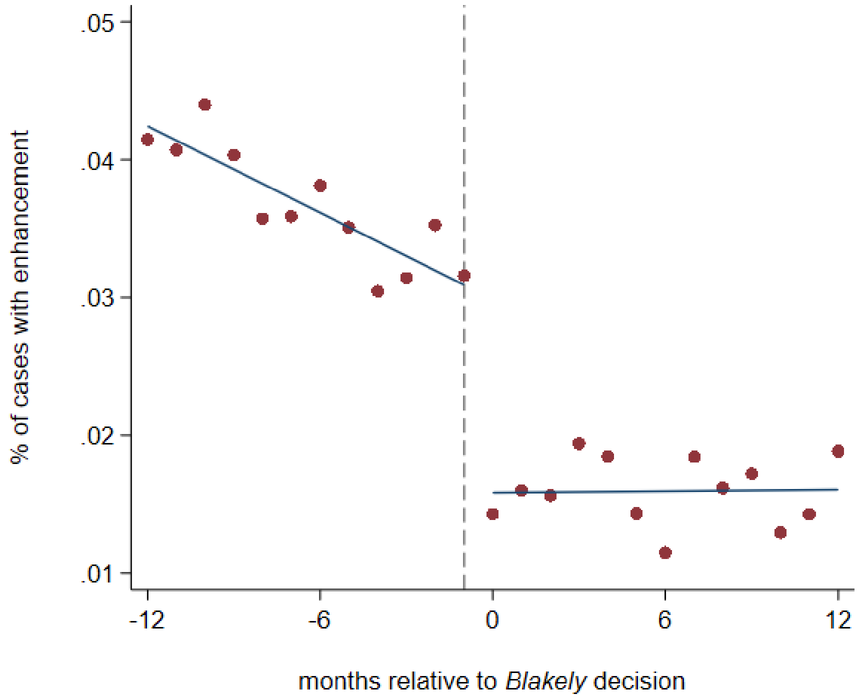


Panel B: *Phat ≥ 0.5 sample*



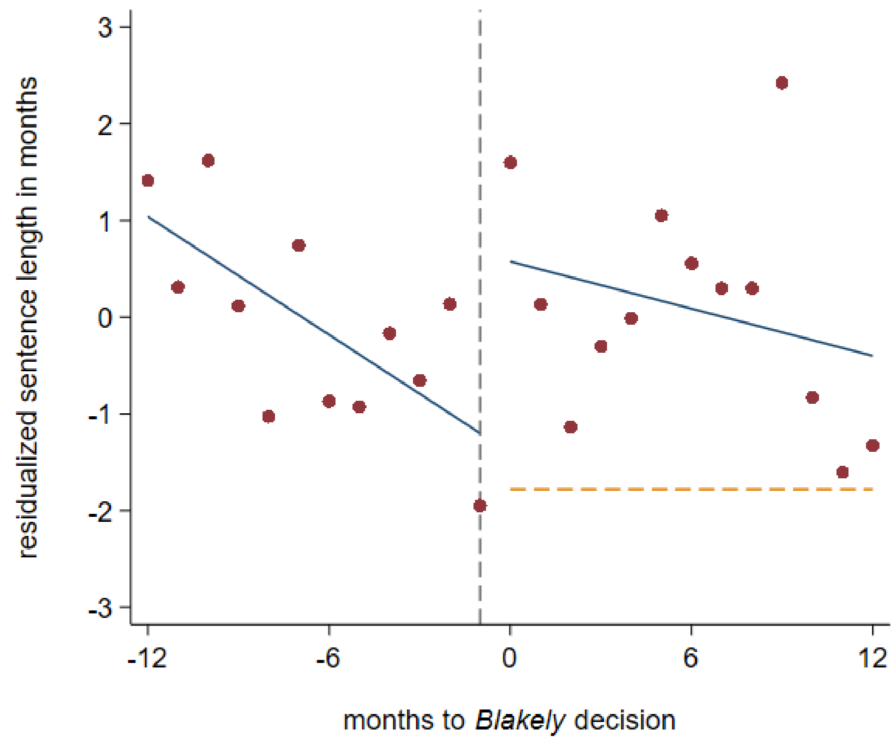
Notes: These graphs illustrate how the aggravated propensity score changes across the timing of *Blakely.* Panel A gives mean propensity scores across time for the full sample while Panel B illustrates the propensity scores for the sample of cases with scores above 0.05. In both cases, *Blakely* has no discernible change in bunching propensities.

Figure 2. Effect of *Blakely* on enhancement



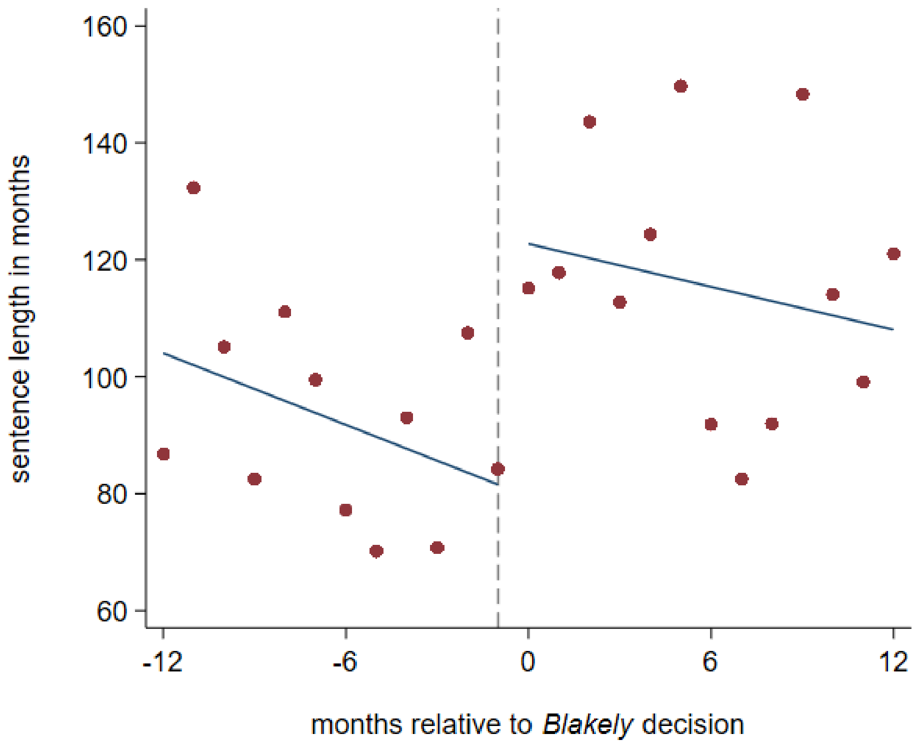
Notes: This graph shows the discontinuity for the percentage of cases with enhancement under a 12-month bandwidth with linear fits for the full sample of cases. The dashed vertical line indicates the timing of the *Blakely* decision.

Figure 3. Effect of *Blakely* on sentence length



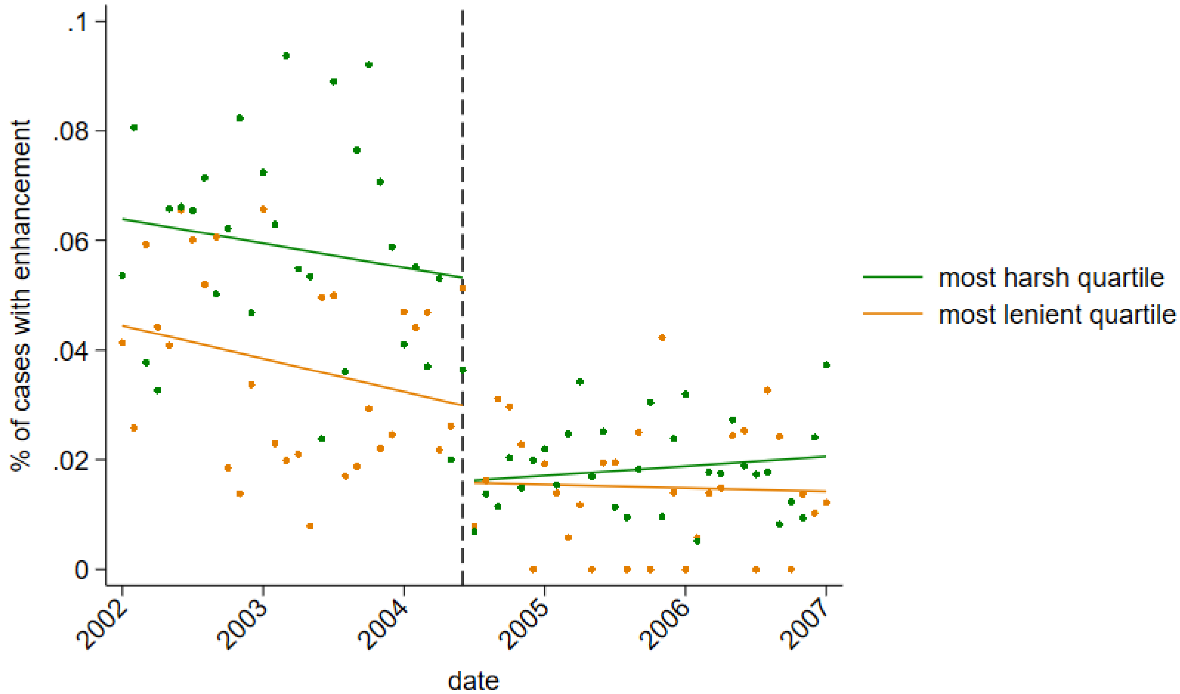
Notes: This graph gives the discontinuity for residualized sentence length using a 12-month bandwidth and linear fit. Sentence length is residualized on grid fixed effects. The vertical dashed line indicates the timing of the *Blakely* decision. The orange, horizontal dashed line gives the predicted decrease in sentence length following *Blakely,* termed the benchmark value. The sample includes cases with propensity scores greater than 0.05.

Figure 4. Effect of *Blakely* on sentence length – jury trial cases



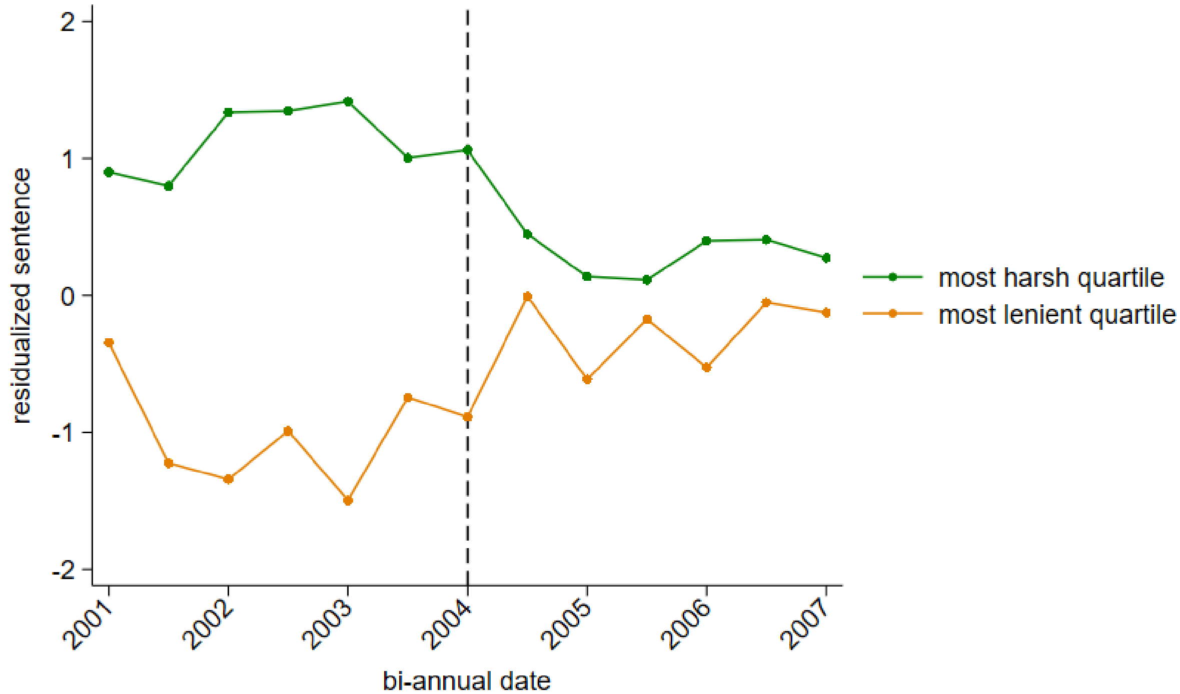
Notes: This graph gives the discontinuity for sentence lengths for cases resolved by jury trial and with a propensity scores greater than 0.05. The fit is linear and bandwidth is 12 months.

Figure 5. Effects of *Blakely* on Enhancement – split by judge leniency



Notes: This graph illustrates heterogeneity in the effects of *Blakely* on enhancement rates by judge type. Specifically, the green points give averages for the harshest quartile of judges while the orange points are for the most lenient quartile. The sample is restricted to cases with a propensity score higher than 0.05.

Figure 6. Effects of *Blakely* on Residual Sentence – split by judge leniency



Notes: This graph gives average residual sentence lengths for the most lenient and harshest judge quartiles. Sentence lengths are residualized on grid and year fixed effects. The sample only includes cases with propensity scores greater than 0.05.

Tables

Table 1. Summary statistics

	full sample		phat ≥0.05 sample	
	mean	sd	mean	sd
Panel A: Defendant Characteristics				
age at charge	30.23	10.19	30.38	11.09
male	0.857	0.350	0.922	0.268
minority race	0.590	0.492	0.611	0.488
private attorney	0.211	0.408	0.168	0.374
public defender	0.259	0.438	0.262	0.440
criminal history points	4.986	5.490	6.025	7.793
Panel B: Case Outcomes				
enhancement	0.027	0.162	0.051	0.220
min sentence length in months	18.50	31.37	38.85	51.78
charge severity	3.609	1.996	5.874	2.282
index crime	0.410	0.492	0.832	0.374
probation length	31.33	10.47	35.13	11.29
incarcerated	0.374	0.484	0.592	0.492
mitigated	0.096	0.295	0.165	0.371
plea bargain	0.979	0.144	0.955	0.208
plead to lower felony	0.144	0.352	0.271	0.445
plead to misdemeanor	0.036	0.186	0.055	0.228
days of case duration	335.6	297.6	359.5	335.5
N	118379		27536	

Notes: Figures are presented for the main analysis sample, which consists of cases with a propensity score over 0.05, and the full sample of cases. Each of these samples contain cases 3 years before or after the timing of *Blakely.* Variables enhancement, male, minority race, private attorney, public defender, index crime, incarcerated, mitigated, plea bargain, plead to lower felony, and plead to misdemeanor are binary. Charge severity is measured on a scale from 1-11, with higher numbers indicating more severe charges.

Table 2. The main effects of *Blakely*

Panel A. The effect of *Blakely* on enhancement rate

	(1)	(2)	(3)	(4)	(5)
Blakely	-0.0200*	-0.0208**	-0.0311***	-0.0140***	-0.0201***
	(0.0119)	(0.0091)	(0.0086)	(0.0017)	(0.0056)
Pre-*Blakely* mean	0.0653	0.0704	0.0670	0.0368	0.0564
p-hat restriction	≥ 0.05	≥ 0.05	≥ 0.05	none	≥ 0.03
bandwidth	9	12	15	12	12
N	6952	9314	11556	39955	16332

*Panel B***.** The effect of enhancement on sentence length

	(1)	(2)	(3)	(4)	(5)
Aggravated	17.33***	17.39***	17.92***	8.82***	13.26***
	(1.846)	(2.016)	(1.751)	(0.871)	(1.547)
Pre-*Blakely* mean	37.55	38.57	38.59	18.55	29.27
p-hat restriction	≥ 0.05	≥ 0.05	≥ 0.05	none	≥ 0.03
Bandwidth	9	12	15	12	12
N	3260	4502	5601	19141	7905

Panel C. The effect of *Blakely* on sentence length

	(1)	(2)	(3)	(4)	(5)
Blakely	1.024	1.996***	1.082	0.493	0.929
	(0.927)	(0.719)	(0.735)	(0.349)	(0.695)
pre-*Blakely* sentence length	37.55	38.57	38.59	18.55	29.27
$\widehat{\Delta}_{blakely}$ (benchmark)	-0.347	-0.362	-0.557	-0.123	-0.267
t-score (benchmark)	1.48	3.28	2.24	1.77	1.72
p-hat restriction	≥ 0.05	≥ 0.05	≥ 0.05	none	≥ 0.03
bandwidth	9	12	15	12	12
N	6952	9314	11556	39955	16332

Notes: This table presents the main three effects as described in Equation 4, with Panel A giving estimates of β_1 in equation 4.1, Panel B giving estimates of δ_1 in Equation 4.2, and Panel C giving estimates of γ_1 in Equation 4.3. $\widehat{\Delta}_{blakely}$ is the predicted, mechanical decrease calculated using the coefficients in Panels A and B. The t-score (benchmark) row gives the t-score value calculated using the benchmark value as the null hypothesis. Across all three panels, standard errors are clustered at the month level, a uniform kernel function is used, and each specification includes grid fixed effects. The p-hat restriction is based on the propensity score described in Section 4.
* p < 0.1, ** p < 0.05, *** p < 0.01

Table 3. Effect of *Blakely* on Charging Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variables:						
charge severity	0.0014	-0.0303	-0.0043	0.0161	0.0182	0.0445
	(0.0544)	(0.0959)	(0.0502)	(0.0868)	(0.0444)	(0.0797)
	{3.597}	{5.910}	{3.600}	{5.917}	{3.597}	{5.900}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
index crime	-0.0206	-0.0278	-0.0175	-0.0165	-0.0148	-0.0147
	(0.0138)	(0.0208)	(0.0126)	(0.0154)	(0.0117)	(0.0140)
	{0.415}	{0.828}	{0.412}	{0.830}	{0.411}	{0.831}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
# of charges (pre-consol.)	-0.150	-0.273	-0.116	-0.211	-0.0284	-0.115
	(0.106)	(0.253)	(0.0926)	(0.186)	(0.111)	(0.157)
	{3.240}	{2.837}	{3.212}	{2.842}	{3.195}	{2.800}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
# of charges (post-consol.)	0.0103	-0.0845	0.0028	-0.0925*	0.0093	-0.0534
	(0.0286)	(0.0568)	(0.0263)	(0.0444)	(0.0259)	(0.0458)
	{1.754}	{1.674}	{1.750}	{1.685}	{1.742}	{1.675}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
dismiss	-0.0045		-0.0033		0.0082	
	(0.0090)		(0.0075)		(0.0062)	
	{0.104}		{0.105}		{0.104}	
	[34519]		[46059]		[57302]	
p-hat restriction	none	≥ 0.05	none	≥ 0.05	none	≥ 0.05
bandwidth	9	9	12	12	15	15

Notes: This table presents results of RD analysis for five different charging outcome dependent variables across six different specifications. The reported effect is the coefficient representing the effect of *Blakely* on each dependent variable. All specifications include criminal history category fixed effects, besides for dismissed cases which do not have that data. Standard errors, clustered at the month level, are presented in parentheses. Pre-*Blakely* means are presented in curly braces. Sample sizes are presented in brackets. The p-hat restriction is based on the propensity score described above. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4. Effect of *Blakely* on Plea Bargain Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variables:						
plea bargain	-0.0016	-0.0104	-0.0053**	-0.0169**	-0.0023	-0.0109
	(0.0019)	(0.0081)	(0.0021)	(0.0079)	(0.0019)	(0.0071)
	{0.979}	{0.952}	{0.979}	{0.953}	{0.979}	{0.954}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
plea to incarceration	0.0159*	-0.0043	0.0103	-0.0130	0.0081	-0.0199
	(0.0089)	(0.0156)	(0.0082)	(0.0152)	(0.0068)	(0.0133)
	{0.355}	{0.545}	{0.358}	{0.550}	{0.357}	{0.546}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
plea to lower felony	0.0222*	0.0511*	0.0201*	0.0359	0.0193**	0.0351
	(0.0109)	(0.0249)	(0.0102)	(0.0239)	(0.0094)	(0.0220)
	{0.142}	{0.276}	{0.144}	{0.273}	{0.144}	{0.272}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
plea to misdemeanor	0.0028	-0.0002	0.0017	-0.0009	0.0038	0.0038
	(0.0051)	(0.0117)	(0.0044)	(0.0102)	(0.0044)	(0.0101)
	{0.036}	{0.054}	{0.036}	{0.053}	{0.036}	{0.053}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
case duration (in days)	-10.71	-26.61**	-15.12*	-23.57**	-9.15	-26.86***
	(7.71)	(9.85)	(7.66)	(9.44)	(6.98)	(8.52)
	{338.9}	{363.6}	{337.1}	{360.3}	{336.1}	{360.2}
	[26656]	[5880]	[35525]	[7909]	[44014]	[9782]
p-hat restriction	none	≥ 0.05	none	≥ 0.05	none	≥ 0.05
bandwidth	9	9	12	12	15	15
N	30002	6952	39955	9314	49699	11556

Notes: This table presents results of RD analysis for five different plea bargain outcome dependent variables across six different specifications. The first four variables are binary while case duration is measured in number of days to resolution. All specifications include grid fixed effects. Standard errors, clustered at the month level, are presented in parentheses. Pre-*Blakely* means are presented in curly braces and sample size in brackets. The p-hat restriction is based on the propensity score described above. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5. Effect of *Blakely* on Sentencing Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variables						
incarceration	0.0190*	0.0081	0.0156*	0.0047	0.0105	-0.0079
	(0.0092)	(0.0154)	(0.0088)	(0.0146)	(0.0074)	(0.0131)
	{0.371}	{0.589}	{0.374}	{0.592}	{0.373}	{0.588}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
probation	1.138***	-0.881	0.741**	-0.781	0.444	-0.660
	(0.265)	(0.564)	(0.275)	(0.558)	(0.292)	(0.498)
	{31.41}	{35.59}	{31.52}	{35.53}	{31.49}	{35.44}
	[18824]	[2848]	[24982]	[3785]	[31143]	[4751]
mitigation	-0.0016	-0.0085	-0.0058	-0.0228*	0.0060	-0.0013
	(0.0050)	(0.0136)	(0.0053)	(0.0126)	(0.0050)	(0.0129)
	{0.096}	{0.167}	{0.096}	{0.166}	{0.095}	{0.163}
	[30002]	[6952]	[39955]	[9314]	[49699]	[11556]
p-hat restriction	none	≥ 0.05	none	≥ 0.05	none	≥ 0.05
bandwidth	9	9	12	12	15	15

Notes: This table presents results of RD analysis for three different sentencing outcome dependent variables across six different specifications. All specifications include grid fixed effects. Standard errors, clustered at the month level, are presented in parentheses. Pre-*Blakely* means are presented in curly braces. Sample sizes are presented in brackets. The p-hat restriction is based on the propensity score described above. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 6. Heterogeneous effects of *Blakely* by judge leniency

	(1) enhancement	(2) enhancement	(3) min. sent	(4) min. sent
Blakely	-0.0238*** (0.0073)	-0.0252*** (0.0073)	2.680*** (0.635)	2.226*** (0.653)
DD_50	0.0103 (0.0083)	0.0101 (0.0084)	-1.043* (0.593)	-0.951 (0.632)
DD_75	-0.0078 (0.0096)	-0.0080 (0.0098)	-2.247*** (0.660)	-2.221*** (0.689)
DD_100	-0.0243*** (0.0090)	-0.0240*** (0.0089)	-4.222*** (0.583)	-4.114*** (0.597)
Fixed effects	yes	yes	yes	yes
Additional controls	no	yes	no	yes
Pre-*Blakely* mean	0.0723	0.0723	39.20	39.20
N	27536	27536	27536	27536

Notes: This table gives the differential effects of *Blakely* on enhancement rate and sentence length across the four judge leniency groups. The sentence length leniency measure is used throughout for these effects. The *Blakely* coefficient gives the effect for the most lenient judges, while DD_50, DD_75, and DD_100 give the difference-in-difference estimates for increasingly strict judges. All specifications include grid and year fixed effects. The additional controls include age, a male dummy, race, defense type, criminal history points, the type of crime, and annual arrest and population rates. Standard errors are clustered at the judicial district level. * p < 0.1, ** p < 0.05, *** p < 0.01